Continued Fractions for Constants and Functions

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Background

Methods

We used the generalized continued fraction definition

$$b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cfrac{a_3}{b_3 + \dots}}} \tag{1}$$

to combine results from various sources of constants and functions expressed in a generalized fraction form. The computation was determined for a depth N that computed the backward recurrence method from the bottom fraction $F_N = \frac{a_N}{b_N}$ up to the top and total fraction $F_0 = b_0 + F_1$ where

$$F_i = \frac{a_i}{b_i + F_{i+1}} \text{ for } i \in \{1, ..., N\}$$

and $F_{N+1} = 0$. All constants, functions, and rules for a_i and b_i including their respective source references can be found in Table 1 of Appendix A.

Results

References

[1] Wikipedia contributors. William brouncker, 2nd viscount brouncker. https://en.wikipedia.org/wiki/William_Brouncker,_2nd_Viscount_Brouncker, August 2026. Accessed: November 8, 2026.

[2] John Baez. Three phases of continued fraction theory. https://golem.ph.utexas.edu/category/2020/09/five_levels_of_continued_fract.html, September 2020.

[3] The Ramanujan Machine. π^2 results. https://ramanujanmachine.com/wp-content/uploads/2021/08/Zeta2_Results.pdf, August 2021.

[4] The Ramanujan Machine. Catalan's constant G results. https://ramanujanmachine.com/wp-content/uploads/2020/06/catalan.pdf, June 2020.

[5] The Ramanujan Machine. Other results. https://ramanujanmachine.com/wp-content/uploads/2020/06/other.pdf, June 2020.

[6] Wikipedia contributors. Euler's continued fraction formula. https://en.wikipedia.org/wiki/Euler%27s_continued_fraction_formula, September 2026. Accessed: November 8, 2026.

[7] Wikipedia contributors. Euler's continued fraction formula. https://en.wikipedia.org/wiki/Euler%27s_continued_fraction_formula, September 2026. Accessed: November 8, 2026.

[8] Wikipedia contributors. Euler's continued fraction formula. https://en.wikipedia.org/wiki/Euler%27s_continued_fraction_formula, September 2026. Accessed: November 8, 2026.

[9] Paramanand Singh. Continued fraction expansion of tan(x). https://paramanands.blogspot.com/2011/04/continued-fraction-expansion-of-tanx.html, April 2011.

[10] Wikipedia contributors. Hyperbolic functions. https://en.wikipedia.org/wiki/Hyperbolic_functions, September 2026. Accessed: November 8, 2026.

Appendix A

All constants, functions, and rules for a_i and b_i including their respective source references can be found in Table 1.

Constant or Function	a_i	b_i
$\frac{\pi}{4}$ [1]	$(2i-3)^2$	$\begin{cases} 0, i = 0 \\ 1, i = 1 \\ 2, i \notin \{0, 1\} \end{cases}$
e [2]	1	$\begin{cases} 2, i = 0 \\ \frac{2i+1}{3}, i \equiv 2 \pmod 3 \\ 1, \text{otherwise} \end{cases}$
$\frac{30}{\pi^2}$ [3]	i^4	$11i^2 + 11i + 3$
$\frac{1}{2G}$ for Catalan's constant G [4]	$-2i^4$	$3i^2 + 3i + 1$
$\frac{1}{1 - \ln 2}$ [5]	$-2i^2(i+1)^2$	$3i^2 + 7i + 4$
e^x [6]	$\begin{cases} 1, i = 1 \\ \frac{-x}{i-1}, i \neq 1 \end{cases}$	$\begin{cases} 0, i = 0 \\ 1, i = 1 \\ 1 + \frac{x}{i-1}, i \notin \{0, 1\} \end{cases}$
$\sin x$ [7]	$\begin{cases} x, i = 1 \\ x^2, i = 2 \\ (2i-4)(2i-3)x^2, i \notin \{1, 2\} \end{cases}$	$\begin{cases} 0, i = 0 \\ 1, i = 1 \\ (2i-2)(2i-1) - x^2, i \notin \{0, 1\} \end{cases}$
$\sinh x$ [8]	$\begin{cases} x, i = 1 \\ -x^2, i = 2 \\ (4-2i)(2i-3)x^2, i \notin \{1, 2\} \end{cases}$	$\begin{cases} 0, i = 0 \\ 1, i = 1 \\ (2i-2)(2i-1) + x^2, i \notin \{0, 1\} \end{cases}$
$\tan x$ [9]	$\begin{cases} x, i = 1 \\ -x^2, i \neq 1 \end{cases}$	$\begin{cases} 0, i = 0 \\ 2i - 1, i \neq 0 \end{cases}$
$\tanh x$ [10]	1	$\begin{cases} 0, i = 0 \\ \frac{2i-1}{x}, i \neq 0 \end{cases}$

Table 1: Generalized Fraction Rules for Constants and Functions